



Department of Electrical and Electronics Engineering

Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. EEE “B” Sec

Course Code & Title: EE8552 Power Electronics

Name of the Faculty member: Mr.R.Thangasankaran

Innovative Practice Description

- **Unit / Topic:** Unit II / Phase Controlled Converters
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO 5
- **Activity Chosen:** Active Learning Method – MATLAB Simulation
- **Justification:**

Rectifiers can be classified as single phase rectifier and three phase rectifier. Single phase rectifiers are classified as 1- Φ half wave and 1- Φ full wave rectifier. Three phase rectifier are classified as 3- Φ half wave rectifier and 3- Φ full wave rectifier. 1- Φ Full wave rectifiers are classified as 1- Φ mid point type and 1- Φ bridge type rectifier. 1- Φ bridge type rectifiers are classified as 1- Φ half controlled and 1- Φ full controlled rectifier. 3- Φ full wave rectifiers are again classified as 3- Φ mid point type and 3- Φ bridge type rectifier. 3- Φ bridge type rectifiers are again divided as 3- Φ half controlled rectifier and 3- Φ full controlled rectifier. After teaching 1- Φ full wave rectifier and 3- Φ full wave rectifier, I had conducted a simulation activity for making the students to understand the operation of both in practical mode and also enhance their learning level and as a teacher I can judge the understanding level of the students.

- **Time Allotted for the Activity:** 30 minutes

- **Details of the Implementation:**

After teaching the concept, gave students five to ten minutes to think about the topic without writing anything.

Total Strength is 29.

Photographer: One student – Mr. M.Senthil Kumar (953620105328)

Reporter: Myself

At the end the Class (Last 10 minutes)

- ✓ I asked the students to simulate the 1- Φ full wave rectifier and 3- Φ full wave rectifier for 10 minutes.



- ✓ Then I asked them to obtain the pulsating output voltage waveform of them for another 5 minutes.
- ✓ Finally, I randomly asked students about the operation of 1- Φ full wave rectifier and 3- Φ full wave rectifier for 5 minutes.

- **CO – PO / PSO mapping:**

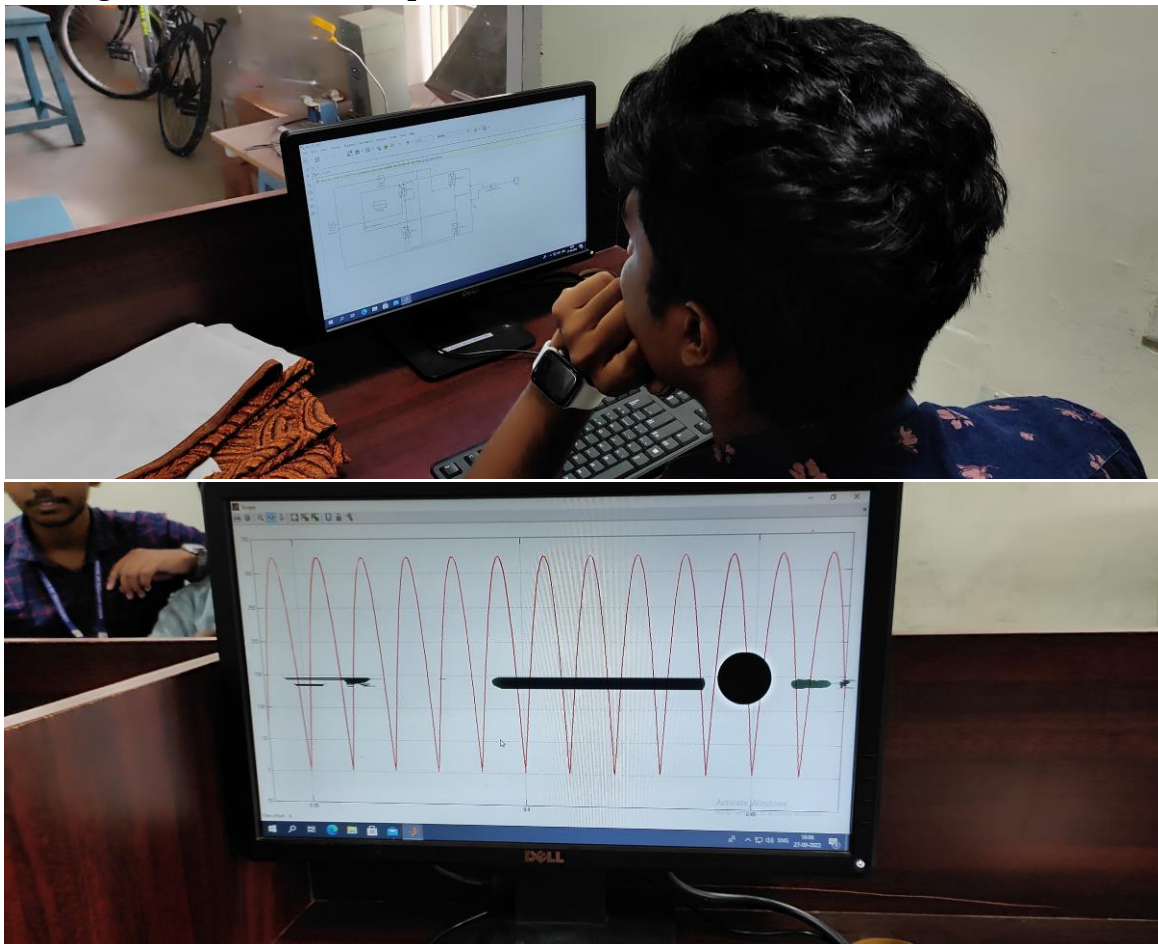
CO	PO 1	PO 2	PO 3	PO 5	PO10	PO12	PSO1	PSO3
CO 2	3	2	1	2	1	1	2	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 7
	2
Justification for correlation	The basic applications of rectifier operating conditions are taught to the students; hence it is moderately correlated (level 2).

- **Images / Screenshot of the practice:**





Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

Students were able to explain the operation of 1- Φ full wave rectifier and 3- Φ full wave rectifier and also they said that they understood the functions of them.

❖ *Benefit of the practice:*

1. Students can able to understand the impact of engineering solution on society
2. Students understood the concept which was reflected from their answers for the questions I have asked.
3. The success of the activity was evaluated by asking the same question during IAT – I – **Around 80% of students answered.**
4. Students can able to explain the concepts in detail in the examination.

❖ *Challenges faced in implementation:*

1. Time utilization for conducting activity, slow learners were not able to answer the question.

References:

- ✓ P.S.Bimbira “Power Electronics” Khanna Publishers, Third Edition, 2003.

Signature of Faculty Member

HOD



Department of Electrical and Electronics Engineering

Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. EEE “B” Sec

Course Code & Title: EE8552 Power Electronics

Name of the Faculty member: Mr.R.Thangasankaran

Innovative Practice Description

- **Unit / Topic:** Unit IV / Inverters
- **Course Outcome:** CO 4
- **Topic Learning Outcome:** TLO 16
- **Activity Chosen:** Active Learning Method – MATLAB Simulation
- **Justification:**

Three Phase Voltage Source Inverters (VSI) is six – step bridge inverter which uses six switches. A step is defined as a change in the firing from one switch to the next switch in the proper sequence. For one cycle of 360° , each step would be of 60° interval for a six – step inverter. The switches would be gated at regular intervals of 60° in proper sequence. There are two possible patterns of gating the switches. In one pattern, each switch conducts for 180° and in the other, each switch conducts for 120° . In both these patterns, gating signals are applied and removed at 60° intervals of the output voltage waveforms. After teaching three phase 180 and 120 Degree mode VSI, I had conducted a simulation activity for making the students to understand the operation of three phase 180 and 120 Degree mode VSI in practical mode and also enhance their learning level and as a teacher I can judge the understanding level of the students.

- **Time Allotted for the Activity:** 30 minutes
- **Details of the Implementation:**

After teaching the concept, gave students five to ten minutes to think about the topic without writing anything.

Total Strength is 29.

Photographer: One student – Mr. S.Vijay Anandh (953620105025)

Reporter: Myself

At the end the Class (Last 10 minutes)

- ✓ I asked the students to simulate the 180 and 120 Degree conduction mode for 10 minutes.



- ✓ Then I asked them to obtain the phase and line voltage waveform differences between 180 and 120 Degree conduction mode for another 5 minutes.
- ✓ Finally, I randomly asked students about the operation of three phase 180 and 120 Degree mode VSI for 5 minutes.

- **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PO 5	PO10	PO12	PSO1	PSO3
CO 4	3	2	1	2	1	1	2	1

(1 – Low

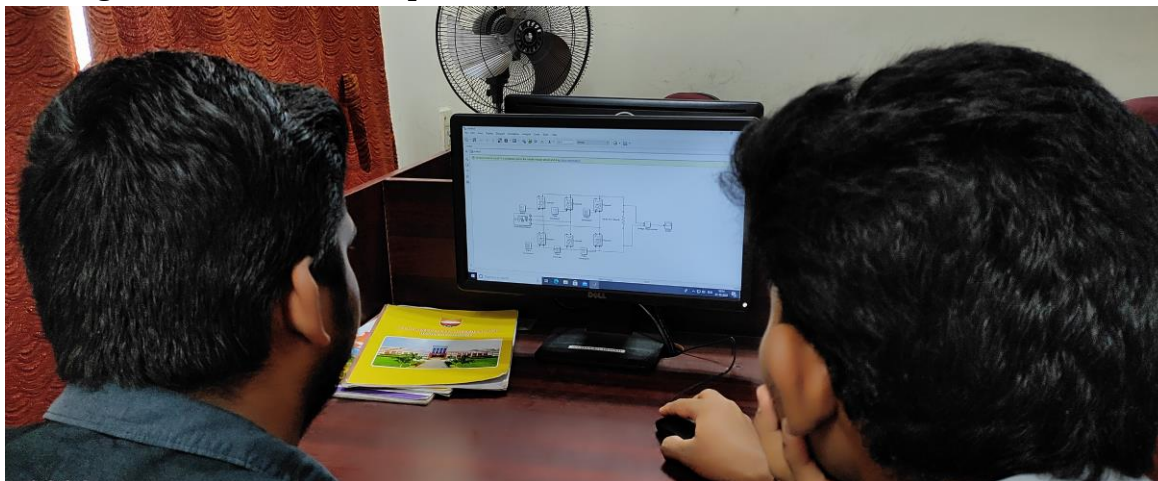
2 – Moderate

3 – High)

- **PO / PSO mapped:**

Innovative practice	PO 7
	1
Justification for correlation	The basic applications of inverter operating conditions are taught to the students, hence it is moderately correlated (level 2).

- **Images / Screenshot of the practice:**





Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

Students were able to classify the different conduction modes of Inverters and also they said that they understood the functions of both 180 and 120 Degree conduction mode.

❖ *Benefit of the practice:*

1. Students can able to understand the impact of engineering solution on society
2. Students understood the concept which was reflected from their answers for the questions I have asked.
3. The success of the activity was evaluated by asking the same question during IAT – III – **Around 90% of students answered.**
4. Students can able to explain the concepts in detail in the examination.

❖ *Challenges faced in implementation:*

1. Time utilization for conducting activity, slow learners were not able to answer the question.

References:

- ✓ P.S.Bimbira “Power Electronics” Khanna Publishers, Third Edition, 2003.

Signature of Faculty Member

HOD